

From: Social Science Quarterly no-reply@wiley.com
Subject: Major revision requested: 7450938
Date: June 2, 2026 at 10:53 AM
To: Matthew C. Nowlin matthew.nowlin@uta.edu

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[External]

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Dear Dr. Matthew C. Nowlin,

Thank you for submitting your manuscript 7450938, titled "The Centrality of Cultural Worldviews in the Climate Change Belief System Network", to Social Science Quarterly.

After careful assessment, our editorial team has decided your manuscript requires major revisions in order to proceed with the peer review process and be considered for publication. Reviewer comments are included at the bottom of this email to assist you in revising your manuscript.

We ask that you return your revised manuscript by August 31, 2026. If you need more time, please let us know. We are happy to discuss an extended deadline.

When submitting your revised manuscript, you will be asked to upload an author response letter that details the revisions you have made in response to editor and reviewer comments. You may also be required to show revisions made within the manuscript files. Please follow any journal-specific guidelines for formatting and submitting revisions outlined in the author guidelines page.

When you have finished revising your manuscript, follow the link below to submit your revision:

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Thank you for submitting your work to Social Science Quarterly. We look forward to receiving your revised manuscript.

Kind regards

The following reviewer comments were taken into consideration during the peer review process.



Reviewer 1 Comments to the Author

The manuscript examines climate change beliefs as an interconnected belief system using a network approach, integrating cultural worldviews, political ideology, environmental orientation, perceived scientific consensus, and policy preferences into a single analytical framework. Building on both hierarchical models of belief systems and recent network approaches, the authors derive three competing hypotheses regarding which types of beliefs should be most central. Using a quota-based U.S. survey from 2017 and Gaussian graphical modeling, the results suggest that egalitarianism, climate risk perceptions, and support for EPA regulation are among the most central nodes, with variation across centrality metrics. The manuscript's main contribution lies in incorporating cultural worldviews into climate belief networks, extending prior work that has largely focused on ideology and partisanship. The discussion argues that different belief types may serve distinct structural functions, thereby bridging hierarchical and network perspectives.

A central issue concerns the treatment of the gateway belief model. The manuscript uses this framework to derive the expectation that perceived scientific consensus should be the most central node in the network, yet the theoretical and empirical implications of the model are not sufficiently developed. The gateway belief model, particularly as elaborated in recent work, is fundamentally concerned with causal pathways and belief-updating processes, often demonstrated through experimental or longitudinal designs (see references below). In contrast, the present study relies on cross-sectional, undirected network data. This creates a conceptual mismatch: centrality in a partial-correlation network does not directly correspond to causal "gateway" status. The current interpretation, that the relatively low centrality of scientific consensus provides limited support for the gateway belief model, is therefore too strong. A more appropriate interpretation, in line with recent literature (see below for references),

is that consensus beliefs may not be structurally central in cross-sectional belief networks, while still functioning as causal entry points under specific informational interventions. The manuscript would benefit from engaging more deeply with recent contributions that refine the gateway belief model, clarify its boundary conditions, and distinguish structural embedding from causal influence. This would substantially strengthen both the theoretical framing and the interpretation of results.

A second major concern is the temporal positioning of the data. The analysis is based on survey data collected in 2017, yet the manuscript situates itself within a rapidly evolving literature on climate belief systems, including studies from the mid-2020s. Given substantial changes in climate discourse, polarization, and policy debates, the authors should more explicitly justify the continued relevance of these data. Without such justification, it remains unclear whether the findings reflect enduring structural properties of belief systems or context-specific patterns tied to an earlier stage of climate communication.

A third issue relates to the interpretation of centrality as indicative of influence. While the manuscript acknowledges the exploratory nature of network analysis, it nonetheless employs language suggesting that certain beliefs “drive” or “shape” others. Given the cross-sectional and undirected nature of the model, such claims are not empirically warranted. This is particularly important in light of the theoretical frameworks invoked, including cultural cognition, solution aversion, and the gateway belief model, all of which make claims about causal processes. The manuscript would benefit from a clearer distinction between structural prominence and causal influence.

Open science practices are insufficiently addressed. There is no indication of preregistration, no a priori power analysis, and no clear statement regarding data and code availability. Given the flexibility inherent in network modeling, particularly decisions regarding node inclusion, scaling, and regularization, transparency is essential. The manuscript should explicitly state whether the analysis was preregistered, provide access to data and code where possible, and include robustness checks or stability analyses beyond the reported bootstrap confidence intervals.

On a more minor level, the literature review is comprehensive but could be more focused, particularly in integrating hierarchical and network perspectives rather than presenting them in parallel. The hypotheses would benefit from greater precision, especially in linking theoretical expectations to specific centrality metrics. The methods section is technically sound but could be streamlined for readability.

section is technically sound but could be streamlined for readability, and the discussion would benefit from a clearer articulation of how the findings advance debates within both policy process research and climate communication.

Overall, the manuscript offers a meaningful extension of climate belief network research by incorporating cultural worldviews. However, it requires substantial revision in its treatment of the gateway belief model, greater caution in interpreting centrality as influence, clearer justification of the data's temporal relevance, and improved transparency regarding analytic decisions.

For strengthening the manuscript, the authors should engage more directly with recent work on the gateway belief model, including studies that examine its causal mechanisms, boundary conditions, and empirical robustness. This literature provides a more nuanced understanding of how consensus beliefs operate and would allow the authors to position their findings as complementary rather than contradictory.

Suggested literature:

Göbel, J. P., Buder, J., Ogiermann, M., Burkhardt, M., Forstner, A., Fischer, H., & Huff, M. (2026). Understanding the gateway:

Unpacking the mechanisms, boundaries, and outcomes of climate consensus messaging. *Journal of Environmental Psychology*, 110, 102913. <https://doi.org/10.1016/j.jenvp.2026.102913>

Said, N., Frauhammer, L. T., & Huff, M. (2022). Pre-registered replication of the gateway belief model – Results from a representative German sample. *Journal of Environmental Psychology*, 84, 101910. <https://doi.org/10.1016/j.jenvp.2022.101910>

Said, N., Frauhammer, L. T., & Huff, M. (2023). Consensus messaging in climate change communication: Metacognition as moderator variable in the gateway belief model. *Journal of Environmental Psychology*, 91, 102128. <https://doi.org/10.1016/j.jenvp.2023.102128>

Reviewer 2 Comments to the Author

Thank you for putting together this paper on such an interesting and important topic. The paper uses original survey data to construct a climate belief network. In doing so, the paper tests which beliefs, drawn from different theoretical perspectives, are most important in the network. In general, I found the paper to be clearly written and the methods to be well done. With that said, there are some areas

the methods to be well done. With that said, there are some areas that I think need to be addressed, which I highlight below.

Most importantly, the impact of the paper is relatively muted. I find it difficult to find the one sentence takeaway from the paper that's particularly convincing and, to some extent, the paper reads as a laundry list of findings. There are 3 hypotheses, many types of beliefs, and three network measures. I think it would benefit the paper to streamline the results and try to provide more substantive interpretations of both the network measures (i.e., can I learn something substantively different from betweenness, closeness, and strength, or should we hope that they all agree) and of what the relatively importance of core beliefs and policy beliefs means for understanding/intervening in the belief network.

Second, much of the theoretical focus of the paper appears to be in adjudicating between a hierarchical and network structure. If this is an important focus of the paper and its impact, I think there needs to be a more substantial discussion of how predictions and interventions would differ between a hierarchical and network structure.

Additionally, I am not convinced that the rejection of the hierarchical structure is not due to the methods that are chosen. The network is undirected and, thus, prevents hierarchy. If you were to use a Bayes net (or some other method that better captures hierarchy), would you reach different conclusions both about the high-level structure and about the relative importance of beliefs? If it is impractical to fit a Bayes net, I would at least like to see a discussion of why the current modeling choices were made and how these choices inform the conclusions. Also on this point, there is a discussion on pg. 5 about fuzziness in hierarchical belief structures and it's unclear to me how that is solved by the network.

Third, the points on scientific consensus appear a bit limited to me. First, the question that is asked about consensus is quite different from the way consensus is operationalized in the literature (where 97% or 99% of scientists are said to agree). This warrants at least some discussion. Second, H2 could use more justification. The GBM is hierarchical, and it would be useful to better translate between that hierarchical structure and the flattened network.

A couple other small points:

- When tuning LASSO, the general approach now is to use cross-validation. The number of nodes and non-zero edges is a bit surprising for LASSO on data of this size. It would be nice to see the robustness to different specifications.

- When comparing the centrality measures across variables, statistical tests should be done.

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